

Question 1

Yr. 11 - Half yearly 2003

(a). Simplify -

(i). $|2| - |-5|$

(ii). $(3x - 6) - 2(5 - 4x)$

(iii). $\sqrt{27} - \sqrt{3} + 3\sqrt{18}$

(b). (i). Evaluate $\frac{143.95 - 11.61}{\sqrt{3.61}}$, giving your answer correct to two decimal places.

(ii). Evaluate $\frac{8.2 \times 10^6}{(3.5 \times 10^8)^2}$, giving your answer in scientific notation correct to two significant figures.

(c). The temperature of a body in F degrees Fahrenheit or C degrees Celsius is related by the formula $F = \frac{9C}{5} + 32$.

(i). Find the value of F when $C = 25$.

(ii). Make C the subject of the formula.

Question 2

(a). Expand and simplify -

(i). $(y + 4)^2$

(ii). $(4x - 3)(2x - 3)$

(b). $g(x) = x^2 + 1$

(i). Evaluate $g(-3)$.

(ii). For what values of x is $g(x) = 2$?

Question 2 (continued)

(c). Solve -

(i). $\frac{x}{3} - 2 = \frac{4x}{5}$

(ii). $x^2 = 3x$

(iii). $2 - 3x < 7$

(d). Factorise completely -

(i). $1 - 8x^3$

(ii). $48x - 3x^3$

(iii). $ab - a + bx - x$

(iv). $5x^2 - 2x - 3$

Question 3

(a). Solve simultaneously -

$$\begin{cases} 3x - 2y = 29 \\ 4x + 5y = 8 \end{cases}$$

(b). Rationalise the denominator and simplify $\frac{8}{\sqrt{5} - 1}$.

(c). Sketch, showing all major features -

(i). $y = 2^{-x}$

(ii). $y = x^3 + 1$

(iii). $y = 25 - x^2$

(d). Solve $|2x + 3| \geq 2$ and graph the solution on the number line.

Question 4

(a). Simplify –

(i). $\frac{x^2 - y^2}{y - x}$

(ii). $\frac{3}{x^2 + 2x - 8} - \frac{2}{x^2 + x - 6}$

(iii). $\frac{3x - 3y}{xy} \div \frac{x^2 - xy}{x^2}$

(b). If $(2 + 5\sqrt{3})^2 = a + \sqrt{b}$, find the values of a and b .

(c). Simplify –

(i). $(-2a^4b)^3$

(ii). $\frac{3c \times 4c^2 \times 5c^3}{3c^2 + 4c^2 + 5c^2}$

(d). Sketch $f(x)$ given that $f(x) = \begin{cases} x^2 & \text{for } x \leq 2 \\ 2x - 5 & \text{for } x > 2 \end{cases}$

Question 5

(a). Write an equation of a straight line which is NOT a function.

(b). Solve $|x - 1| = 2x - 1$.

(c). Express $0.1\dot{2}\dot{4}$ as a simple fraction. Show working.

Question 5 (continued)

(d). State the domain for the following –

(i). $y = x + 2$

(ii). $y = \frac{3}{x}$

(iii). $x^2 + y^2 = 100$

(e). State the range for the following –

(i). $y = |x|$

(ii). $y = -\sqrt{9 - x^2}$

(iii). $y = 2^x - 8$

Question 6

(a). Sketch the region satisfied by all three inequalities –

$$x^2 + y^2 \leq 25, 2x - y > 0 \text{ and } y \geq 0.$$

(b). A cyclist rides for 5 hours at a certain speed and then for 4 hours at a speed 6 km/h greater than her original speed. If she rides 655.8 km altogether, what was –

(i). her average speed?

(ii). her initial speed?

(c). For the graph $y = 16 - 2(x + 5)^3$, find –

(i). the x -intercept;

(ii). the y -intercept.

Question 6 (continued)

- (d). The lines $x = 3$, $y = -2$ and $y = 10$ are tangents to a circle which never crosses the y -axis. Find its --
- (i). centre;
 - (ii). radius.
- (e). With full working to support your answer, determine whether the function $f(x) = \frac{4x}{x^2 + 4}$ is odd, even or neither.
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Question 7

- (a). Solve $3x^2 = 7x - 1$, leaving your answer in exact form.
- (b). Sketch, showing all major features --
- (i). $y = -\sqrt{4 - x^2}$
 - (ii). $y = 5 - \frac{3}{x + 2}$
 - (iii). $y = |3x - 4|$
- (c). Express $\frac{1 - x^{-1}}{x^{\frac{1}{2}}}$ in simplest form (without fractional/negative indices) and with rational denominator. [Assume x is an integer.]
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Question 8

- (a). Find values for A , B and C if $y^2 + 16y + 94 - 6x$ is expressed in the form $(y + C)^2 - B(x + A)$.
- (b). If $x + y = 1$ and $x^3 + y^3 = 19$, find the value of $x^2 + y^2$.
- (c). Sketch $x \geq \frac{1}{y}$.
- (d). For every pair of numbers a and b , the function $f(x)$ satisfies $b^2 f(a) = a^2 f(b)$. If $f(2) \neq 0$, find the value of $\frac{f(5) - f(1)}{f(2)}$.
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Question 1

Total
= 113

(a). (i). -3 ①

(ii). $3x - 6 - 10 + 8x = 11x - 16$ ②

(iii). $3\sqrt{3} - \sqrt{3} + 9\sqrt{2} = 2\sqrt{3} + 9\sqrt{2}$ ②

(b). (i). 69.65 ② [1 for answer, 1 for 2 d.p.]

(ii). $6 \cdot 7 \times 10^{-11}$ ② [1 for answer in SH, 1 for 2 sig. fig.]

(c). (i). $F = 77$ ①

(ii). $F = \frac{9c}{5} + 32$

$5F = 9c + 160$

$c = \frac{5F - 160}{9}$ ②

Total 12

Question 2

(a). (i). $y^2 + 8y + 16$ ①

(ii). $8x^2 - 6x - 12x + 9$
 $= 8x^2 - 18x + 9$ ②

(b). (i). $g(-3) = 10$ ①

(ii). $x^2 + 1 = 2$
 $x = \pm 1$ ②

(c). (i). $\frac{x-6}{3} = \frac{4x}{5}$

$5x - 30 = 12x$

$7x = -30 \therefore x = -4\frac{2}{7}$ ③

(ii). $x^2 - 3x = 0$

$x(x-3) = 0 \therefore x = 0 \text{ or } 3$ ②

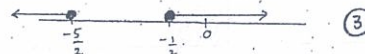
(iii). $-3x < 5$

$x > -\frac{5}{3}$ ②

3. (d).

$2x + 3 \geq 2 \text{ or } 2x + 3 \leq -2$

$x \geq -\frac{1}{2} \text{ or } x \leq -\frac{5}{2}$



Total 17

Question 4

(a). (i). $\frac{(x+y)(x-y)}{-(x-y)} = -x-y$ ②

(ii). $\frac{8}{(x+4)(x-2)} - \frac{2}{(x+3)(x-2)}$

$= \frac{3x+9 - (2x+8)}{(x+4)(x-2)(x+3)}$

$= \frac{x+1}{(x+4)(x-2)(x+3)}$ ③

(iii). $\frac{3(x-y)}{xy} \times \frac{x^2}{x(x-y)}$
 $= \frac{3}{y}$ ②

(b). $4 + 20\sqrt{3} + 75 = 79 + \sqrt{1200}$

$\therefore a = 79 \quad b = 1200$ ②

2 (continued)

(a). (i). $(1-2x)(1+2x+4x^2)$ ①

(ii). $3x(16-x^2)$

$= 3x(4+x)(4-x)$ ②

(iii). $a(b-1) + x(b-1)$
 $= (a+x)(b-1)$ ②

(iv). $(5x+3)(x-1)$ ②

Total 20

Question 3

(a). $15x - 10y = 145$ ③
 $8x + 10y = 16$ ③
 $\therefore x = 7, y = -4$ ③

(b). $\frac{8}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} = \frac{8\sqrt{5}+8}{5-1} = \frac{8\sqrt{5}+8}{4}$
 $= 2\sqrt{5}+2$ ③

(c). (i). ② [1 for shape, 1 for y-int.]

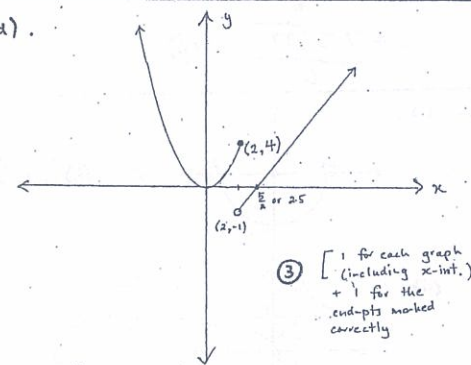
(ii). ③ [1 for shape, 2 for both intercepts]

(iii). ③ [1 for shape, 1 for all x and y intercepts]

4. (c). (i). $-8a^{12}b^5$ ①

(ii). $\frac{60c^6}{12c^2} = 5c^4$ ②

(d).



Total 15

Question 5

(a). $x = 2 \text{ or } x = -2 \text{ etc.}$ ①

(b). $x-1 = 2x-1 \text{ or } x-1 = 1-2x$
 $0 = x \quad 3x = 2$

$\therefore x = 0 \text{ or } \frac{2}{3}$

Check $x = 0$
 $LHS = 0-1 = -1$
 $RHS = 0-1 = -1 \quad x$

Check $x = \frac{2}{3}$
 $LHS = \left|\frac{2}{3} - 1\right| = \frac{1}{3}$
 $RHS = 2 \times \frac{2}{3} - 1 = \frac{1}{3} \checkmark$

$\therefore x = \frac{2}{3} \text{ only solution}$ ③

(c). let $x = 0.1242424 \dots$

$100x = 12.4242424 \dots$

$99x = 12.3 \quad x = \frac{41}{330}$ ③

3. (d). (i). $x \in \mathbb{R}$

(ii). $x \in \mathbb{R}$ except $x=0$

(iii). $-10 \leq x \leq 10$ (3)

(e). (i). $y \geq 0$

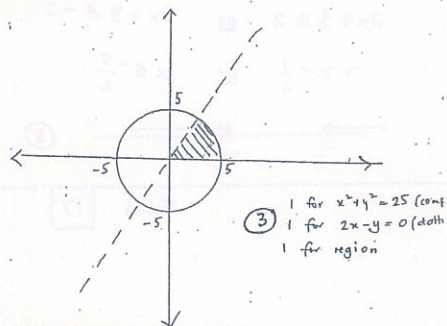
(ii). $-3 \leq y \leq 0$

(iii). $y > -8$ (3)

Total 13

Question 6

(a).



(b). (i). 72.87 km/h (1)

(ii). let $x = \text{initial speed}$

$$5x + 4(x+6) = 655.8$$

$$9x = 631.8$$

$$\therefore x = 70.2 \text{ km/h} \leftarrow \text{initial speed}$$

(c). (i). $0 = 16 - 2(x+5)^3$ $(x+5)^3 = 8$
 $\therefore x = -3$ (2)
 $(-3, 0)$

(ii). $y = -234$ (1) $(0, -234)$

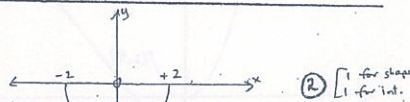
Question 7

(a). $3x^2 - 7x + 1 = 0$

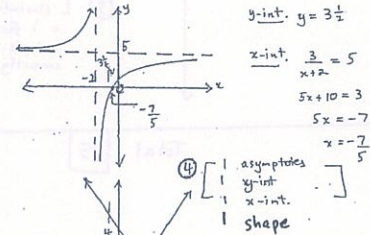
$$x = \frac{7 \pm \sqrt{49 - 4 \times 3}}{6}$$

$$= \frac{7 \pm \sqrt{37}}{6}$$

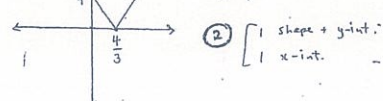
(b). (i).



(ii).



(iii).



7. (c).

$$\frac{1 - \frac{1}{x}}{\sqrt{x}}$$

$$= \frac{x-1}{x\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{x}}$$

$$= \frac{x\sqrt{x} - \sqrt{x}}{x^2}$$

Total 13

Question 8

(a). $y^2 + 16y + 94 - 6x$

$$y^2 + 16y + 64 + 30 - 6x$$

$$= (y+8)^2 - 6(x-5)$$

$$A = -5 \quad B = 6 \quad C = 8$$

(b). $x^3 + y^3 = (x+y)(x^2 - xy + y^2)$

$$19 = x^2 + y^2 - xy$$

$$\therefore 19 = x^2 + (1-x)^2 - x(1-x)$$

$$19 = 3x^2 - 3x + 1$$

$$3x^2 - 3x - 18 = 0$$

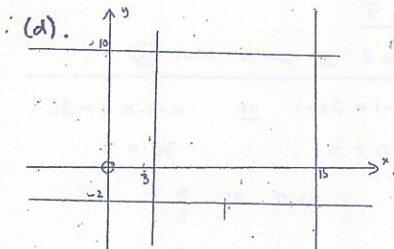
$$x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$x = 3 \text{ or } -2$$

$$y = -2 \text{ or } 3$$

$$\therefore x^2 + y^2 = 9 + 4 = 13$$



(i). $(9, 4)$

(ii). $r = 6$ (2)

(e). $f(x) = \frac{4x}{x^2 + 4}$

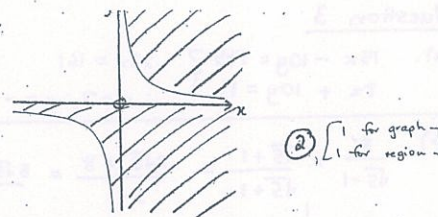
$$f(-x) = \frac{4(-x)}{(-x)^2 + 4} = \frac{-4x}{x^2 + 4}$$

$$-f(x) = \frac{-4x}{x^2 + 4}$$

$$\therefore f(-x) = -f(x) \quad \therefore \text{odd function}$$

Total 13

8. (c).



(d). $\frac{f(a)}{f(b)} = \frac{a^2}{b^2}$

$$\frac{f(5) - f(1)}{f(2)} = \frac{25}{4} - \frac{1}{4}$$

$$= \frac{24}{4} = 6$$

Total 10